

7	(a)	(i)	Battery capacity $= (42.0 \times 10^3)(3600)$ [C1] $= 1.51 \times 10^8$ [A1] SI base unit: $\text{kg m}^2 \text{s}^{-2}$ [B1]	
		(ii)	To improve distance travelled on a single charge (or any other sensible answer) [B1]	
	(b)	(i)	The specific energy of a battery is the amount of energy it is <u>able to provide (or store)</u> per unit mass. [B1] Note: Do not accept “energy per unit mass”	
		(ii)	Mass of battery $= (42.0 \times 10^3) / 275$ [C1] $= 153 \text{ kg}$ [A1]	
		(iii)	Lower the centre of gravity [B1] to improve stability [B1]	
	(c)	(i)	$\text{volume} = (33.0 \times 10^3) / (0.200) = 165 \times 10^3 \text{ cm}^3$ [A1]	
		(ii)	Correct plot [B1] Suitable straight line of best fit. [B1]	
		(iii)	$2.6 \% \text{ of } 1.0 \times 10^7 = 2.6 \times 10^5 \text{ cm}^3$ [M1] From graph, battery volume will reach this size by 2023 (or 2023.5). [A1]	

	(d)	(i)	Number of cells required = $414 / 3.6 = 115$ [B1] Connected in series [B1]	
		(ii)	Temperature can be maintained within working temperature. [B1] Prevent overheating which can cause damage to batteries / fire. [B1]	
	(e)		Maximum electrical power = $(175)(2500) = 437.5$ kW [C1] Maximum propulsion power = $0.72 (437.5) = 315$ kW [A1]	